The double commutator of the CT expansion

$\frac{1}{2}[[\hat{F},\hat{A}],\hat{A}]$, and the eight blocks

This is the last piece of the first-order structure: the second-order term of the CT expansion, in the convention of `ct_F_A.tex` and `ct_V_A.tex` (crossed Fock order, excitations, general indices over $\{i\}\cup\{a\}$). With

$$\hat{A}=\hat{T}-\hat{T}^\dagger,\qquad \hat{T}=\hat{T}_1+\hat{T}_2,\qquad \hat{C}=[\hat{F},\hat{A}], \tag{1}$$

the second-order term is

$$\tfrac{1}{2}[[\hat{F},\hat{A}],\hat{A}]=\tfrac{1}{2}[\hat{C},\hat{A}]=\tfrac{1}{2}[\hat{C},\hat{T}]+\tfrac{1}{2}[\hat{C},\hat{T}]^\dagger, \tag{2}$$

the second equality because $\hat{A}$ is anti-Hermitian, which makes $\hat{C}$ Hermitian, so that $[\hat{C},\hat{T}]^\dagger=-[\hat{C},\hat{T}^\dagger]$ and $[\hat{C},\hat{A}]=[\hat{C},\hat{T}]-[\hat{C},\hat{T}^\dagger]$. Only the $\hat{T}$ half has to be computed; the other half is its conjugate.

1. Splitting $\hat{C}$

Write the two halves of the Fock commutator separately,

$$\hat{C}_1=[\hat{F},\hat{T}_1],\qquad \hat{C}_2=[\hat{F},\hat{T}_2],\qquad \hat{C}=\hat{C}_1+\hat{C}_2+\hat{C}_1^\dagger+\hat{C}_2^\dagger, \tag{3}$$

with $\hat{C}_r^\dagger=-[\hat{F},\hat{T}_r^\dagger]$ the conjugate of $\hat{C}_r$. Both $\hat{C}_1$ and $\hat{C}_2$ are already in closed form in `ct_F_A.tex`: two terms for the one-body block, four for the two-body block.

Since the commutator is linear in its first argument,

$$[\hat{C},\hat{T}]=[\hat{C}_1,\hat{T}]+[\hat{C}_2,\hat{T}]+[\hat{C}_1^\dagger,\hat{T}]+[\hat{C}_2^\dagger,\hat{T}], \tag{4}$$

and expanding each $[\cdot,\hat{T}]=[\cdot,\hat{T}_1]+[\cdot,\hat{T}_2]$ gives the eight commutators of the sheets:

$$\boxed{\begin{aligned}\tfrac{1}{2}[[\hat{F},\hat{A}],\hat{A}]=\tfrac{1}{2}\Big\{&[\hat{C}_1,\hat{T}_1]+[\hat{C}_1,\hat{T}_2]+[\hat{C}_2,\hat{T}_1]+[\hat{C}_2,\hat{T}_2]\\&+[\hat{C}_1^\dagger,\hat{T}_1]+[\hat{C}_1^\dagger,\hat{T}_2]+[\hat{C}_2^\dagger,\hat{T}_1]+[\hat{C}_2^\dagger,\hat{T}_2]\Big\}+\text{h.c.}.\end{aligned}} \tag{5}$$

2. What the eight blocks are

The first four of (5) are the nested double commutators of the Fock operator with two $\hat{T}$'s,

$$[\hat{C}_r,\hat{T}_s]=[[\hat{F},\hat{T}_r],\hat{T}_s],\qquad r,s\in\{1,2\}, \tag{6}$$

and the last four are the same with the conjugate inner commutator, $[\hat{C}_r^\dagger,\hat{T}_s]$. They are not related to each other by conjugation in a way that saves work:

$$[\hat{C}_r,\hat{T}_s]^\dagger=-[\hat{C}_r^\dagger,\hat{T}_s^\dagger], \tag{7}$$

which is a statement about $[\cdot, \hat{T}_s^\dagger]$, not about $[\cdot, \hat{T}_s]$, so each of the eight has to be evaluated. In particular the two mixed nested commutators are different operators,

$$\big[[\hat{F}, \hat{T}_1], \hat{T}_2\big] \neq \big[[\hat{F}, \hat{T}_2], \hat{T}_1\big], \tag{8}$$

since $\hat{T}_1\hat{T}_2 - \hat{T}_2\hat{T}_1$ does not vanish (four terms), so the Jacobi shortcut is not available and both mixed blocks are kept.

Each block is evaluated the same way: commute once to get $\hat{C}_r$ (or $\hat{C}_r^\dagger$) from `ct_F_A.tex`, then commute the result with $\hat{T}_s$ using the $(1,1)$ or $(1,2)$ rule. Counting terms before merging:

block	inner object	inner terms	terms out
$[\hat{C}_1, \hat{T}_1] = \big[[\hat{F}, \hat{T}_1], \hat{T}_1\big]$	$\hat{C}_1$ (1 body)	2	$2 \times 2 = 4$
$[\hat{C}_1, \hat{T}_2] = \big[[\hat{F}, \hat{T}_1], \hat{T}_2\big]$	$\hat{C}_1$ (1 body)	2	$2 \times 4 = 8$
$[\hat{C}_2, \hat{T}_1] = \big[[\hat{F}, \hat{T}_2], \hat{T}_1\big]$	$\hat{C}_2$ (2 body)	4	$4 \times 2 = 8$
$[\hat{C}_2, \hat{T}_2] = \big[[\hat{F}, \hat{T}_2], \hat{T}_2\big]$	$\hat{C}_2$ (2 body)	4	$4 \times 4 = 16$
the four conjugate-inner blocks	$\hat{C}_r^\dagger$	same	same

Every one of these terms is kept: as in the first-order blocks, the terms of a block are different operators with different index patterns, and the collection into $\bar{h}$ and $\bar{g}$ reads a coefficient off each of them.

2.1 A worked block: $\big[[\hat{F}, \hat{T}_1], \hat{T}_1\big]$

The simplest of the eight, and the one that answers the natural question of whether it vanishes. With $[\hat{F}, \hat{T}_1]$ from `ct_F_A.tex` and $\hat{T}_1 = t_j^b \hat{E}_j^b$ with b virtual and j occupied,

$$\big[[\hat{F}, \hat{T}_1], \hat{T}_1\big] = f_q^a \, t_i^a \, t_j^\beta \, [\hat{E}_i^q, \hat{E}_j^\beta] - f_i^p \, t_i^a \, t_j^\beta \, [\hat{E}_p^a, \hat{E}_j^\beta], \tag{9}$$

and the $(1,1)$ rule gives

$$[\hat{E}_i^q, \hat{E}_j^\beta] = \delta_i^\beta \hat{E}_j^q - \delta_j^q \hat{E}_i^\beta, \qquad [\hat{E}_p^a, \hat{E}_j^\beta] = \delta_p^\beta \hat{E}_j^a - \delta_j^a \hat{E}_p^\beta. \tag{10}$$

Two of the four resulting terms vanish identically, and not by cancellation: δ_i^β ties an occupied label to a virtual one, and so does δ_j^a, so both are zero whatever the amplitudes are. The other two survive and are equal to each other under $i \leftrightarrow j$, $a \leftrightarrow \beta$,

$$\big[[\hat{F}, \hat{T}_1], \hat{T}_1\big] = -f_j^a \, t_i^a \, t_j^\beta \, \hat{E}_i^\beta - f_i^\beta \, t_i^a \, t_j^\beta \, \hat{E}_j^a = -2 \, f_j^a \, t_i^a \, t_j^\beta \, \hat{E}_i^\beta. \tag{11}$$

So the block does *not* vanish. $[\hat{F}, \hat{T}_1]$ is a one-body operator of the same shape as $\hat{T}_1$ but with different coefficients, and two one-body operators do not commute merely because they look alike. Checked on matrices: the four delta terms of (10) reproduce the block to 2.2×10^{-15}, and the block is non-zero.

3. The check

The identities of §1 are operator identities, so they can be checked on explicit matrices with no reference to the $\hat{E}$ algebra. `tests/test_ct_structure.py` does this with random Hermitian $\hat{F}$ and arbitrary $\hat{T}_1, \hat{T}_2$; the values are

identity	$\max\lvert\Delta\rvert$
$\hat{C} = [\hat{F}, \hat{A}] = \hat{C}_1 + \hat{C}_2 + \hat{C}_1^\dagger + \hat{C}_2^\dagger$	3.7×10^{-14}
$[\hat{C}, \hat{T}]$ equals the four commutators of (4)	6.8×10^{-13}
$[\hat{C}, \hat{T}]$ equals the eight of (5)	7.6×10^{-13}
$\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}] = \frac{1}{2}([\hat{C}, \hat{T}] + \text{h.c.})$	4.8×10^{-13}
$[\hat{C}_1, \hat{T}_1] = \big[[\hat{F}, \hat{T}_1], \hat{T}_1\big]$	0.0
$[\hat{C}_1, \hat{T}_2]^\dagger = -[\hat{C}_1^\dagger, \hat{T}_2^\dagger]$	1.7×10^{-13}

and the check also asserts that $[[\hat{F}, \hat{T}_1], \hat{T}_2]$ and $[[\hat{F}, \hat{T}_2], \hat{T}_1]$ differ, so the mixed blocks cannot be taken as equal.

4. Where this leaves the derivation

With (5) the second-order term is reduced to the eight blocks above, each of which is two applications of rules that are already verified. What remains is not algebra of this kind but two other things:

- **the three-body reduction.** The $(2, 2)$-derived blocks produce three-body operators, which the $[\dots]_{1,2}$ truncation folds back into one- and two-body form with the extended normal ordering of Kutzelnigg and Mukherjee — an approximation, and one that has to be recorded as such wherever the result is used;
- **the collection.** Reading $\bar{h}$ and $\bar{g}$ off the assembled monomials, term by term, in the general basis $C_p^q \hat{E}_p^q + \frac{1}{2} C_{pq}^{rs} \hat{E}_{pq}^{rs}$ of `ct_F_A.tex`, which is where the index patterns of the previous notes have to be preserved rather than cancelled.

References

[1] W. Kutzelnigg and D. Mukherjee, *Normal order and extended Wick theorem for a multiconfiguration reference wave function*, J. Chem. Phys. **107**, 432 (1997).

[2] T. Yanai and T. Shiozaki, *Canonical transcorrelated theory with projected Slater-type geminals*, J. Chem. Phys. **136**, 084107 (2012).